

Week 9 · Class 1: Practice Exercises — ANSWER KEY

Statistical Tests & Confidence Intervals

2025-12-31

Segment A — Statistical Tests

Non-AI Exercises

1. Understanding Standard Error

1.1 Conceptual Questions

1. What is standard error?

Answer: **b** (The standard deviation of the sampling distribution)

2. What happens to standard error when you double your sample size?

Answer: **c** (It decreases by $\sqrt{2}$, approximately 1.41)

3. Which factor does NOT affect the standard error?

Answer: **b** (Population mean)

1.2 Calculating Standard Error

Given a sample with $n = 36$ and $s = 12$:

$$SE = 12 / \sqrt{36} = 2$$

1.3 True or False: Standard Error Concepts

T - Larger samples always have smaller standard errors **T** - Standard error measures the precision of your estimate **T** - To halve the SE, you need $4\times$ the sample size **T** - SE depends on both sample size and variability **F** - Standard error is the same as standard deviation

2. The t-Distribution

2.1 Multiple Choice: t-Distribution Basics

1. The t-distribution was developed to solve what problem?

Answer: **b** (Dealing with uncertainty when estimating standard deviation from small samples)

2. As sample size increases, the t-distribution:

Answer: **b** (Approaches the normal distribution)

3. For a sample of $n = 10$, the degrees of freedom are:

Answer: **b** (9)

2.2 Conceptual Understanding

Why does the t-distribution have heavier tails than the normal distribution?

With small samples, we're less certain about the true variability. The heavier tails account for the additional uncertainty from estimating the standard deviation from the sample.

3. One-Sample t-Tests

3.1 Hypothesis Testing Setup

Null Hypothesis (H_0): $\mu = 50\%$ (The average approval rating equals 50%)

Alternative Hypothesis (H_a): $\mu \neq 50\%$ (The average approval rating differs from 50%)

Test statistic formula: $t = (\bar{x} - \mu_0) / SE = (\bar{x} - \mu_0) / (s / \sqrt{n})$

3.2 Practice Calculation

Step 1: Calculate SE

$$SE = 10 / 5 = 2$$

Step 2: Calculate t-statistic

$$t = (47.5 - 50) / 2 = -1.25$$

Step 3: Conclusion

Is $|t| > 2.064$? **No** ($1.25 < 2.064$)

Conclusion: **Fail to reject H_0 . The approval rating is not significantly different from 50% at $\alpha = 0.05$.**

3.3 Interpreting p-values

If $p = 0.08$, what does this mean?

Answer: **b** (If the null is true, there's an 8% chance of seeing this result or more extreme)

4. Two-Sample t-Tests

4.1 Match: Research Questions to Comparisons

a = **3** (Two political parties) b = **1** (Treatment vs. Control groups) c = **4** (Battleground vs. Safe states) d = **2** (Two different age groups)

4.2 Understanding the SE of Differences

F - The SE of difference equals the average of the two group SEs **T** - Larger samples lead to smaller SE of difference **T** - We add variances, not standard errors **T** - Equal sample sizes minimize the SE of difference

4.3 Practice Two-Sample Test

Step 1: Calculate the difference in means

$$\text{Difference} = 72 - 65 = 7$$

Step 2: Calculate SE of difference

$$\text{SE_diff} = \sqrt{(144/40 + 225/40)} = \sqrt{(9.225)} = 3.04$$

Step 3: Calculate t-statistic

$$t = 7 / 3.04 = 2.30$$

Step 4: Interpret

Significant? **Yes**

Why? $|t| = 2.30 > 2.00$, so we reject H_0 at $\alpha = 0.05$

4.4 When t-Tests Can Go Wrong

1. Answer: **b**
2. Answer: **b**
3. True or False:

T - With $n > 30$ per group, the t-test usually performs well even when the data are skewed

T - Extreme outliers can distort both the means and the variances a t-test relies on **T** - The classical (non-Welch) t-test assumes the two groups have equal variances **F** - A t-test requires the outcome to be perfectly normally distributed in every case

5. Standard Error in Regression

5.1 Multiple Choice: Regression SE

1. The standard error of a regression coefficient tells us:

Answer: **a** (How much the coefficient would vary across repeated samples)

2. Which increases the SE of a regression coefficient?

Answer: **b** (More variability in the outcome - larger residuals)

3. A coefficient is 0.08 with SE = 0.02. Is it significant at $\alpha = 0.05$?

Answer: **a** (Yes, because $|0.08 / 0.02| = 4 > 1.96$)

5.2 Reading Regression Output

1. For every \$1,000 increase in `ad_spending`, vote share increases by: 0.035 percentage points (or 35 percentage points per \$1 million)

2. The incumbent coefficient SE is 1.200. What does this mean?

If we repeated the study many times, the incumbent coefficient would typically vary by about ± 1.2 percentage points around the estimated value of 8.5.

3. Which predictor has the most precise estimate (smallest SE relative to estimate)?

`ad_spending`: $0.035/0.008 = 4.38$ `incumbent`: $8.500/1.200 = 7.08$

Answer: incumbent (larger t-statistic means more precise relative to effect size)

5.3 Coefficient Plot Interpretation

1. Which coefficients are statistically significant?

Education only (CI does not include zero)

2. How can you tell from the confidence intervals?

If the 95% CI excludes zero, the coefficient is significant at $\alpha = 0.05$. Education's CI (0.30 to 0.70) excludes zero, while Age (-0.03 to 0.33) and Income (-0.30 to 0.20) both include zero.

3. Which estimate is most precise?

Education

Why? It has the smallest SE relative to its estimate (SE = 0.10 vs. estimate = 0.50)

6. t-Tests in Regression Context

6.1 True or False: Understanding Regression t-Tests

T - Every regression coefficient is automatically tested against zero **T** - The t-statistic is calculated as: coefficient / SE **T** - In multiple regression, t-tests are conditional on other variables **T** - A two-sample t-test is equivalent to regression with a binary predictor **F** - Larger t-statistics indicate weaker evidence

6.2 Conditional Tests

1. Why did the coefficient change from 0.045 to 0.035?

Campaign spending was correlated with incumbency. When we control for incumbency, we isolate the effect of spending alone, which is smaller because some of the original effect was confounded with being an incumbent.

2. What does the t-test mean in the simple regression?

It tests whether campaign spending predicts vote share (without controlling for anything else).

3. What does the t-test mean in the multiple regression?

It tests whether campaign spending predicts vote share **after controlling for (holding constant) incumbency status**.

6.3 Match: Interpretation of t-Statistics

a = **3** (Weak evidence, fail to reject null) b = **2** (Moderate evidence against null, significant) c = **4** (Very weak evidence, fail to reject null) d = **1** (Strong evidence against null, significant)

7. Degrees of Freedom

7.1 Calculating Degrees of Freedom

1. One-sample t-test with $n = 50$: $df = \mathbf{49}$
2. Two-sample t-test with $n_1 = 30$, $n_2 = 40$: $df = \mathbf{68}$ (or use Welch's approximation)
3. Regression with $n = 100$ and 3 predictors: $df = \mathbf{96}$ ($n - k - 1 = 100 - 3 - 1$)

7.2 Multiple Choice: DF Concepts

Why do we subtract 1 for degrees of freedom in a one-sample test?

Answer: **b** (Because we use up one df estimating the mean)

AI Exercises

Tips for Working with Claude:

- Ask for **R code using only tidyverse** (no other packages)
- Request **simple, focused answers** to your specific question—not complex analyses
- Ask Claude to **explain what the code is doing** since you're learning
- Avoid asking for visualizations or plots in these exercises
- Include the output of `glimpse()` in your prompt so Claude knows your variable names

Example prompt: “Using tidyverse in R, compare turnout rates between treatment and control groups and calculate the standard error of the difference. Keep the code simple and explain what each line does. Here is what my data looks like: [paste glimpse output]”

8. Get-Out-The-Vote Analysis

```
# Load data
gotv_data <- read_csv("data/gotv_experiment.csv")

# 1. Summary statistics by group
gotv_summary <- gotv_data %>%
  group_by(treatment) %>%
  summarise(
    n = n(),
    mean_turnout = mean(turnout_pct),
    sd_turnout = sd(turnout_pct),
    .groups = "drop"
  )
gotv_summary
```

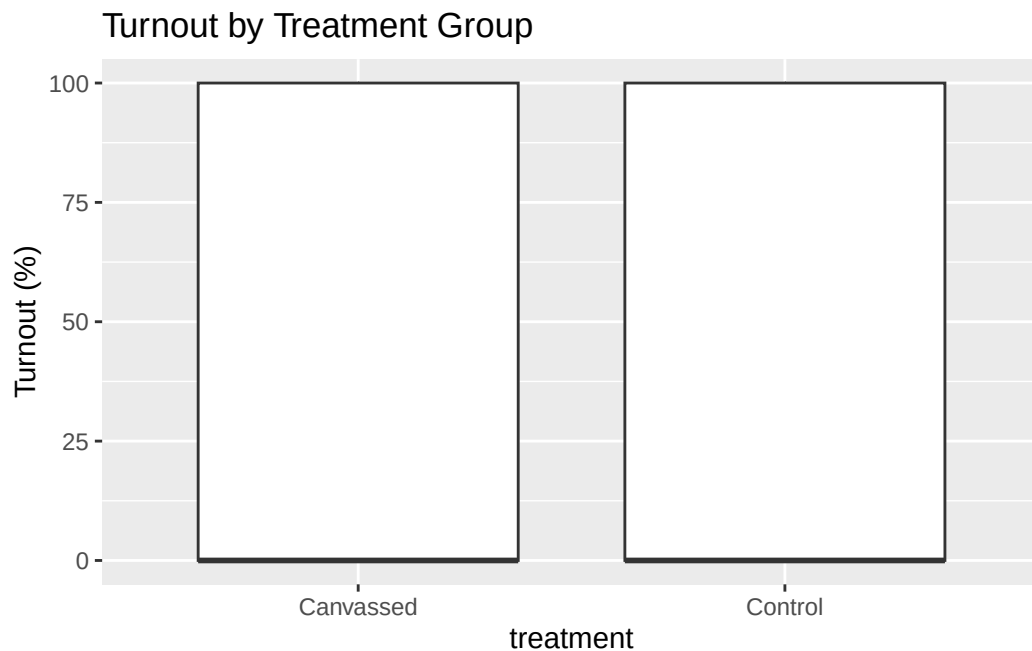
```
# A tibble: 2 x 4
  treatment      n mean_turnout sd_turnout
  <chr>      <int>      <dbl>      <dbl>
1 Canvassed   250         48.8         50.1
2 Control     250         44.4         49.8
```

```
# 2. Two-sample t-test
gotv_test <- t.test(turnout_pct ~ treatment, data = gotv_data)
gotv_test
```

Welch Two Sample t-test

```
data: turnout_pct by treatment
t = 0.98514, df = 497.98, p-value = 0.325
alternative hypothesis: true difference in means between group Canvassed and group Control is
95 percent confidence interval:
 -4.375281 13.175281
sample estimates:
mean in group Canvassed    mean in group Control
                48.8                44.4
```

```
# 3. Visualization
ggplot(gotv_data, aes(x = treatment, y = turnout_pct)) +
  geom_boxplot() +
  labs(title = "Turnout by Treatment Group",
       y = "Turnout (%)")
```



```
# 4. Effect size (canvassed minus control)
gotv_effect <- gotv_summary$mean_turnout[gotv_summary$treatment == "Canvassed"] -
  gotv_summary$mean_turnout[gotv_summary$treatment == "Control"]
gotv_effect
```


[1] 4.4

Interpretation: The canvassed group turned out 4.4 percentage points higher, but the difference is not statistically significant ($t = 0.99$, $p = 0.325$; 95% CI -4.4 to 13.2). The interval includes zero but also includes effects large enough to matter, so the correct conclusion is that this study is too imprecise to settle the question — not that canvassing has no effect.

9. Political Knowledge Study

```
# Load data
knowledge_data <- read_csv("data/political_knowledge.csv")

# 1. Test if news watchers have higher knowledge
knowledge_test <- t.test(knowledge_score ~ watches_news, data = knowledge_data)
knowledge_test
```

Welch Two Sample t-test

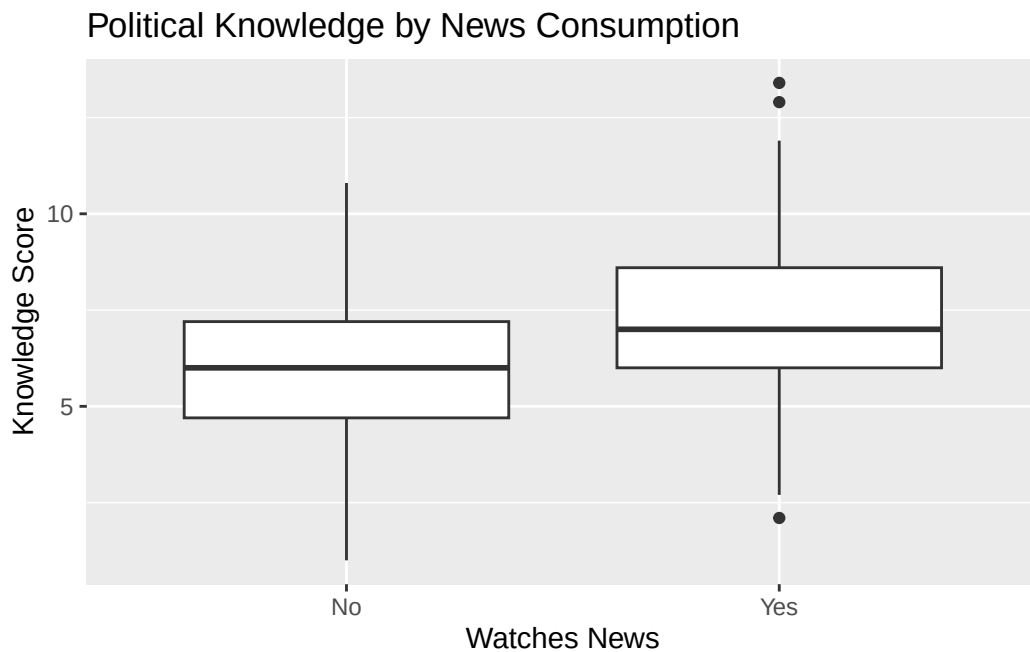
```
data: knowledge_score by watches_news
t = -4.1348, df = 184.15, p-value = 5.393e-05
alternative hypothesis: true difference in means between group No and group Yes is not equal
95 percent confidence interval:
 -1.8518024 -0.6554501
sample estimates:
mean in group No mean in group Yes
    5.948673      7.202299
```

```
# 2. Summary statistics with SE
knowledge_summary <- knowledge_data %>%
  group_by(watches_news) %>%
  summarise(
    n = n(),
    mean = mean(knowledge_score),
    sd = sd(knowledge_score),
    se = sd / sqrt(n),
    .groups = "drop"
  )
knowledge_summary
```

```
# A tibble: 2 x 5
  watches_news    n mean   sd   se
  <chr>         <int> <dbl> <dbl> <dbl>
1 No           113  5.95  2.11 0.199
2 Yes            87  7.20  2.14 0.229

# 3. Calculate effect size (watchers minus non-watchers)
knowledge_effect <- knowledge_summary$mean[knowledge_summary$watches_news == "Yes"] -
  knowledge_summary$mean[knowledge_summary$watches_news == "No"]

# 4. Visualization
ggplot(knowledge_data, aes(x = watches_news, y = knowledge_score)) +
  geom_boxplot() +
  labs(title = "Political Knowledge by News Consumption",
       x = "Watches News", y = "Knowledge Score")
```



Interpretation: News watchers score 1.25 points higher on average ($t = 4.13$, $p < 0.001$). However, this is observational data, so we cannot conclude causation. Potential confounders include education, age, and political interest.

10. Campaign Spending Regression

```
# Load data
campaign_data <- read_csv("data/campaign_spending.csv")

# 1. Simple regression
model_simple <- lm(vote_share ~ campaign_spending, data = campaign_data)
summary(model_simple)
```

Call:

```
lm(formula = vote_share ~ campaign_spending, data = campaign_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-18.661	-4.224	0.125	3.910	20.958

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	36.471877	1.722494	21.17	< 2e-16 ***
campaign_spending	0.025455	0.003264	7.80	3.48e-13 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.878 on 198 degrees of freedom

Multiple R-squared: 0.235, Adjusted R-squared: 0.2312

F-statistic: 60.83 on 1 and 198 DF, p-value: 3.48e-13

```
# 2. Multiple regression
model_multiple <- lm(vote_share ~ campaign_spending + incumbent,
                     data = campaign_data)
summary(model_multiple)
```

Call:

```
lm(formula = vote_share ~ campaign_spending + incumbent, data = campaign_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-16.2351	-3.9639	0.0171	4.1618	15.6923

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	34.907540	1.496271	23.330	< 2e-16 ***
campaign_spending	0.024178	0.002817	8.584	2.75e-15 ***
incumbent	7.756377	0.929824	8.342	1.26e-14 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.927 on 197 degrees of freedom

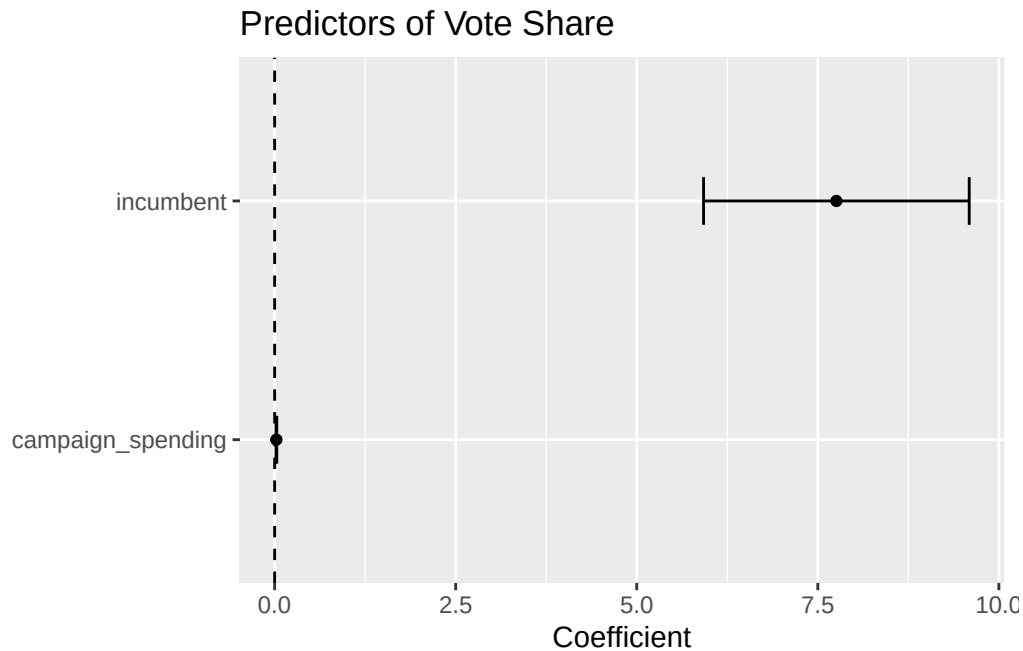
Multiple R-squared: 0.4347, Adjusted R-squared: 0.429

F-statistic: 75.75 on 2 and 197 DF, p-value: < 2.2e-16

```
# 3. Extract coefficients using broom
simple_tidy <- tidy(model_simple)
multiple_tidy <- tidy(model_multiple)

simple_spending <- simple_tidy %>% filter(term == "campaign_spending")
multiple_spending <- multiple_tidy %>% filter(term == "campaign_spending")

# 4. Coefficient plot
tidy(model_multiple, conf.int = TRUE) %>%
  filter(term != "(Intercept)") %>%
  ggplot(aes(x = estimate, y = term)) +
  geom_vline(xintercept = 0, linetype = "dashed") +
  geom_point() +
  geom_errorbarh(aes(xmin = conf.low, xmax = conf.high), height = 0.2) +
  labs(title = "Predictors of Vote Share",
       x = "Coefficient", y = "")
```



Interpretation:

- In the simple model, spending coefficient = 0.025 ($t = 7.8$, $p < 0.001$)
- In the multiple model, spending coefficient = 0.024 ($t = 8.6$, $p < 0.001$)
- The coefficient decreased because spending is correlated with incumbency
- Controlling for incumbency, each \$1,000 in spending increases vote share by 0.024 percentage points

Segment B — Confidence Intervals

Non-AI Exercises

1. Understanding Confidence Intervals

1.1 Conceptual Understanding

1. What does a 95% confidence interval tell us?

Answer: **c** (If we repeated the study 100 times, about 95 CIs would contain the true parameter)

2. A 95% CI for a coefficient is [2.1, 5.8]. What can we conclude?

Answer: **b** (The effect is statistically significant at $\alpha = 0.05$, because the CI excludes zero)

3. Which statement about confidence intervals is TRUE?

Answer: **c** (Larger samples generally produce narrower CIs)

1.2 True or False: CI Concepts

T - A 99% CI is wider than a 95% CI for the same data **T** - If a 95% CI excludes zero, then $p < 0.05$ **T** - The confidence interval is random, but the parameter is fixed **F** - Narrower CIs always mean larger effect sizes **T** - Sample size affects the width of confidence intervals

2. Calculating Confidence Intervals

2.1 Manual CI Calculation

Step 1: Calculate the margin of error

$$\text{Margin} = 2.00 \times 0.12 = \mathbf{0.24}$$

Step 2: Calculate CI bounds

$$\text{Lower bound} = 0.45 - 0.24 = \mathbf{0.21} \quad \text{Upper bound} = 0.45 + 0.24 = \mathbf{0.69}$$

95% CI: [0.21, 0.69]

2.2 Interpreting Different Confidence Levels

90% CI: Lower = 0.25, Upper = 0.65, Width = **0.40**

95% CI: Lower = 0.21, Upper = 0.69, Width = **0.48**

99% CI: Lower = 0.13, Upper = 0.77, Width = **0.64**

Question: What happens to the width as confidence increases?

The interval gets wider - we need a wider range to be more confident.

2.3 CI and Significance

1. 95% CI: [0.2, 0.8]

Significant? **Yes** Why? **The CI excludes zero**

2. 95% CI: [-0.1, 0.5]

Significant? **No** Why? **The CI includes zero**

3. 95% CI: [-2.5, -0.8]

Significant? **Yes** Why? **The CI excludes zero (all values are negative)**

3. Three Roads to the Same Conclusion

3.1 Connecting SE, t-statistic, p-value, and CI

Step 1: Calculate t-statistic

$$t = 0.035 / 0.008 = 4.375$$

Step 2: With $df = 100$, is $|t| > 1.98$?

Yes, so the coefficient is **significant**

Step 3: Calculate 95% CI

$$CI = 0.035 \pm (1.98 \times 0.008) = 0.035 \pm 0.0158 = [0.019, 0.051]$$

Step 4: Does the CI exclude zero?

Yes, confirming the coefficient is **significant**

3.2 Comparing Two Studies

1. Which study has a more precise estimate?

Study B (large sample)

2. Why?

Study B has a much smaller SE (0.09 vs 0.30) and narrower CI (width = 0.36 vs 1.22), indicating higher precision.

3. Both studies found similar effect sizes. Why does Study B have a smaller p-value?

Larger sample size leads to smaller SE, which produces a larger t-statistic (8.67 vs 2.50) and thus smaller p-value.

3.3 Match: Statistical Concepts

a = **3** (Measure of sampling variability/precision) b = **1** (Coefficient is 3.2 standard errors from zero) c = **4** (Probability of seeing this result if null is true) d = **2** (Range of plausible values for the parameter)

4. Sample Size and Precision

4.1 True or False: Sample Size Effects

F - Doubling sample size doubles precision (it reduces SE by $\sqrt{2}$, not doubles precision) **T** - Larger samples produce narrower confidence intervals **T** - With $n = 10,000$, even tiny effects can be statistically significant **F** - Sample size affects the point estimate (sample size affects precision, not the estimate) **T** - Critical values get smaller as sample size increases

4.2 Practical Implications

1. Is this result statistically significant at $\alpha = 0.05$?

Yes ($p = 0.048 < 0.05$)

2. Is the confidence interval informative? Why or why not?

Not very informative - the CI is very wide (width = 10.2), ranging from nearly zero to over 10. This indicates high uncertainty about the true effect size.

3. What would you recommend for a follow-up study?

Increase the sample size substantially to get a more precise estimate with a narrower CI. The current pilot suggests a possible effect, but we need more data to pin down the magnitude.

5. Writing About Confidence Intervals

5.1 Identify Good vs. Bad Writing

Write-up A: B (Bad) Problem: Numbers without interpretation, no substantive meaning, no discussion of practical significance

Write-up B: G (Good) Strengths: Complete statistics, substantive interpretation, discusses both statistical and practical significance

Write-up C: B (Bad) Problem: Incorrect interpretation - misunderstands what confidence intervals mean (the parameter is fixed, not probabilistic)

Write-up D: B (Bad) Problem: Confuses “not significant” with “no effect,” ignores the point estimate and CI

5.2 Fix the Bad Write-Ups

Original (D): “Education had no effect ($p = 0.12$).”

Improved: “Education showed a positive but non-significant relationship with income ($\beta = 2.5$, 95% CI: $[-0.7, 5.7]$, $t(148) = 1.56$, $p = 0.12$). While not statistically significant, the confidence interval is wide and includes effect sizes that would be practically meaningful. A larger study would be needed to estimate this relationship more precisely.”

6. Critical Values and qt()

6.1 Confidence Level Practice

a = 3 (0.95) b = 2 (0.975) c = 1 (0.995) d = 4 (0.9995)

Why these values?

For a two-tailed test, we split α equally between both tails. For a 95% CI, $\alpha = 0.05$, so we have 0.025 in each tail. We want the value that has 97.5% of the distribution below it (0.975).

7. Common Pitfalls

7.1 Statistical vs. Practical Significance

1. Is this statistically significant?

Yes ($p < 0.001$)

2. Is this practically significant? Why or why not?

Not very practical - The effect size is only 0.08 points on a 0-100 scale (less than 0.1%), which is negligible in practical terms.

3. What’s the problem with focusing only on the p-value?

With a very large sample ($n = 50,000$), even tiny, practically meaningless effects become statistically significant. The p-value tells us the effect is real, but not whether it matters.

7.2 Non-Significant No Effect

1. What's wrong with this interpretation?

It confuses “failure to reject the null” with “proof of no effect.” The wide CI shows we have high uncertainty, not evidence of zero effect.

2. What does the CI tell us?

The CI [-2.0, 15.0] includes both negative and positive values, including some that would be large and meaningful. We simply lack precision to pin down the true effect.

3. Write a better interpretation:

“The effect was positive but not statistically significant ($\hat{\mu} = 6.5$, 95% CI: [-2.0, 15.0], $p = 0.13$). The confidence interval is very wide and includes both small negative effects and large positive effects. This suggests the study was underpowered rather than demonstrating no effect exists. A larger study is needed.”

7.3 Multiple Testing

1. Answer: **b** ($20 \times 0.05 = 1$)
2. Answer: **b**
3. Report all 20 tests rather than only the significant one; pre-register the analysis; or apply a multiple-comparison correction such as Bonferroni.

7.4 Match: Pitfalls to Solutions

a = **1** (Always report and discuss effect sizes) b = **3** (Acknowledge uncertainty; consider power)
c = **4** (Comment on CI width/precision) d = **2** (Check for confounders; need experimental design)

AI Exercises

Tips for Working with Claude:

- Ask for **R code using only tidyverse** (no other packages)
- Request **simple, focused answers** to your specific question—not complex analyses
- Ask Claude to **explain what the code is doing** since you're learning
- Avoid asking for visualizations or plots in these exercises
- Include the output of `glimpse()` in your prompt so Claude knows your variable names

Example prompt: “Using tidyverse in R, run a regression and calculate the 95% confidence interval for the coefficient. Keep the code simple and explain what each line does. Here is what my data looks like: [paste glimpse output]”

8. Education and Test Scores

```
# Load data
education_data <- read_csv("data/education_spending.csv")

# Fit model
model_edu <- lm(test_score ~ spending_per_student, data = education_data)

# 1. Examine output
summary(model_edu)
```

Call:

```
lm(formula = test_score ~ spending_per_student, data = education_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-15.2338	-5.4363	-0.7018	4.6481	26.2830

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	6.584e+01	3.581e+00	18.386	< 2e-16 ***
spending_per_student	1.362e-03	2.849e-04	4.779	6.19e-06 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7.762 on 98 degrees of freedom

Multiple R-squared: 0.189, Adjusted R-squared: 0.1807

F-statistic: 22.84 on 1 and 98 DF, p-value: 6.189e-06

```
# 2. Calculate 95% CIs
confint(model_edu)
```

	2.5 %	97.5 %
(Intercept)	58.733710901	72.946159919
spending_per_student	0.000796155	0.001926812

```
# 3. Interpret
tidy(model_edu, conf.int = TRUE)
```

```
# A tibble: 2 x 7
  term                estimate std.error statistic  p.value  conf.low conf.high
<chr>                <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
1 (Intercept)         65.8      3.58     18.4 1.56e-33  58.7     72.9
2 spending_per_student 0.00136 0.000285    4.78 6.19e- 6  0.000796  0.00193
```

Interpretation: The coefficient is 0.00136 per dollar (95% CI: [0.00080, 0.00193], $t(98) = 4.78$, $p < 0.001$). Scaled to something interpretable, each additional **\$1,000** in per-student spending is associated with a 1.36-point increase in test scores, with a plausible range of 0.8 to 1.93 points. The effect is clearly distinguishable from zero, but note the width: the upper end is more than twice the lower end, so the estimate is significant without being especially precise.

9. Campaign Spending Analysis

```
# Load data
campaign_data <- read_csv("data/campaign_spending.csv")

# Fit model
model_campaign <- lm(vote_share ~ campaign_spending + incumbent,
                     data = campaign_data)

# Get CIs
confint(model_campaign)
```

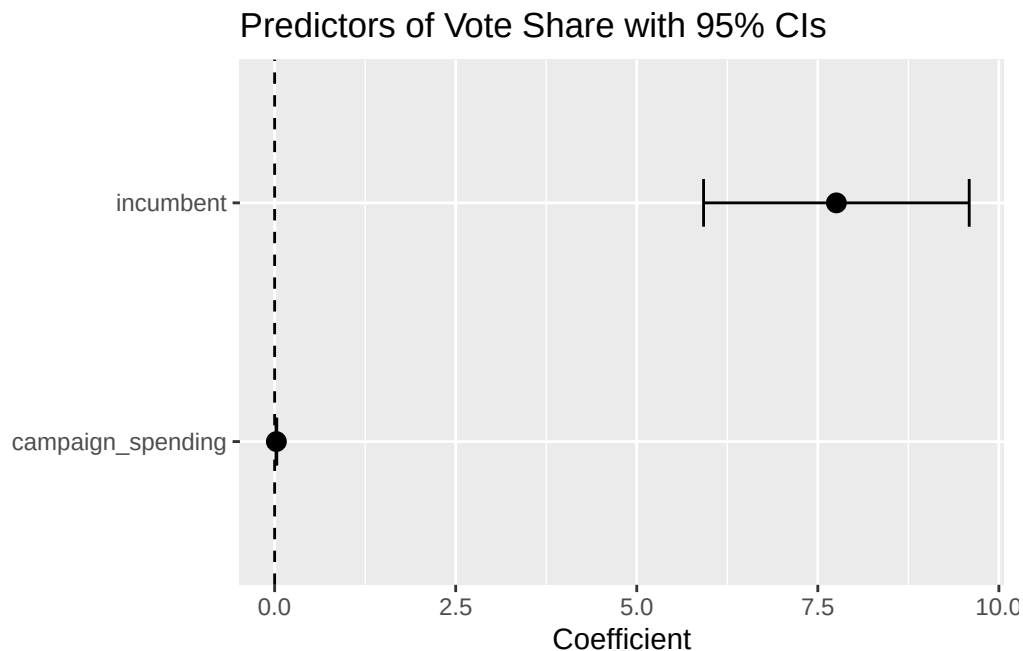
```
                2.5 %      97.5 %
(Intercept)    31.95677417 37.85830525
campaign_spending 0.01862336 0.02973318
incumbent       5.92268995 9.59006430
```

```
tidy(model_campaign, conf.int = TRUE)
```

```
# A tibble: 3 x 7
  term                estimate std.error statistic  p.value  conf.low conf.high
<chr>                <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
```

1 (Intercept)	34.9	1.50	23.3	1.36e-58	32.0	37.9
2 campaign_spending	0.0242	0.00282	8.58	2.75e-15	0.0186	0.0297
3 incumbent	7.76	0.930	8.34	1.26e-14	5.92	9.59

```
# Coefficient plot
tidy(model_campaign, conf.int = TRUE) %>%
  filter(term != "(Intercept)") %>%
  ggplot(aes(x = estimate, y = term)) +
  geom_vline(xintercept = 0, linetype = "dashed") +
  geom_point(size = 3) +
  geom_errorbarh(aes(xmin = conf.low, xmax = conf.high), height = 0.2) +
  labs(title = "Predictors of Vote Share with 95% CIs",
       x = "Coefficient", y = "")
```



Interpretation:

- **Campaign spending:** = 0.0242 (95% CI: [0.0186, 0.0297], $p < 0.001$). Spending is measured in thousands of dollars, so each additional \$1,000 is associated with a 0.024-point gain in vote share — statistically clear, but it would take roughly \$41,000 of additional spending to move vote share by a single point.
- **Incumbent:** = 7.76 (95% CI: [5.92, 9.59], $p < 0.001$). Incumbents receive about 7.8 percentage points more than challengers, holding spending constant. This is the larger and more practically important of the two effects.

10. Comparing Sample Sizes

```
# Load data
voter_study_small <- read_csv("data/voter_turnout_small.csv")
voter_study_large <- read_csv("data/voter_turnout_large.csv")

# Fit models
model_small <- lm(turnout_pct ~ canvassing_contacts, data = voter_study_small)
model_large <- lm(turnout_pct ~ canvassing_contacts, data = voter_study_large)

# Compare
tidy(model_small, conf.int = TRUE)
```

```
# A tibble: 2 x 7
```

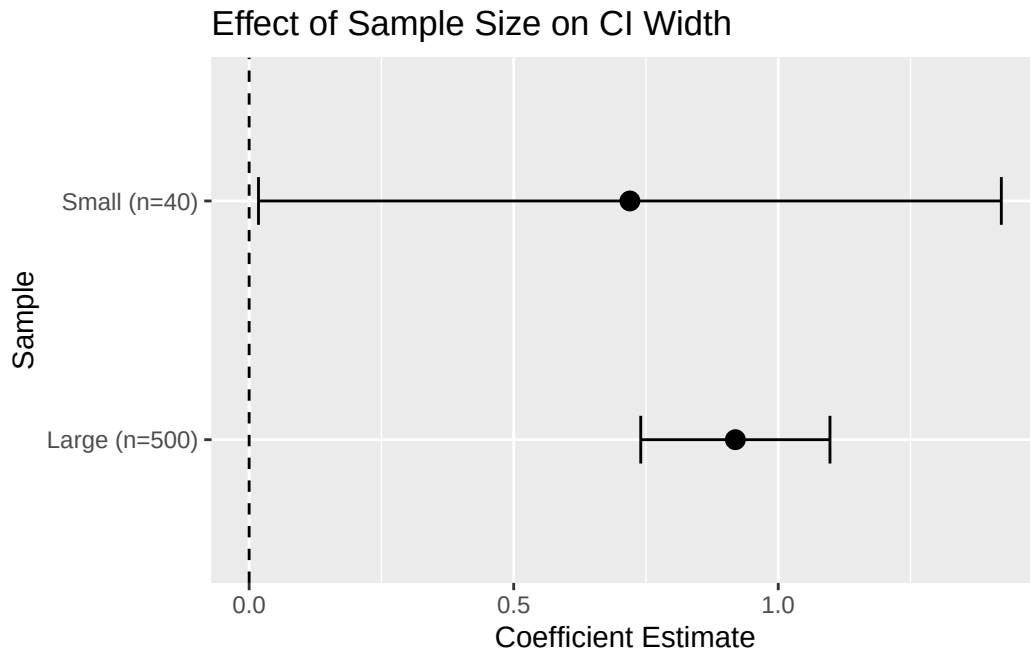
term	estimate	std.error	statistic	p.value	conf.low	conf.high
<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1 (Intercept)	45.1	5.63	8.01	1.10e-9	33.7	56.5
2 canvassing_contacts	0.720	0.347	2.08	4.48e-2	0.0177	1.42

```
tidy(model_large, conf.int = TRUE)
```

```
# A tibble: 2 x 7
```

term	estimate	std.error	statistic	p.value	conf.low	conf.high
<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1 (Intercept)	43.7	1.43	30.6	1.86e-116	40.9	46.5
2 canvassing_contacts	0.919	0.0910	10.1	6.19e- 22	0.740	1.10

```
# Compare CIs visually
bind_rows(
  tidy(model_small, conf.int = TRUE) %>% mutate(sample = "Small (n=40)"),
  tidy(model_large, conf.int = TRUE) %>% mutate(sample = "Large (n=500)")
) %>%
  filter(term == "canvassing_contacts") %>%
  ggplot(aes(x = estimate, y = sample)) +
  geom_vline(xintercept = 0, linetype = "dashed") +
  geom_point(size = 3) +
  geom_errorbarh(aes(xmin = conf.low, xmax = conf.high), height = 0.2) +
  labs(title = "Effect of Sample Size on CI Width",
       x = "Coefficient Estimate", y = "Sample")
```



Comparison:

- Small sample ($n = 40$): $\hat{\beta} = 0.72$, $SE = 0.35$, 95% CI $[0.02, 1.42]$, width 1.4
- Large sample ($n = 500$): $\hat{\beta} = 0.92$, $SE = 0.09$, 95% CI $[0.74, 1.1]$, width 0.36
- **The large-sample CI is about 3.9× narrower.** Both studies estimate the same underlying slope, but the small study's interval barely excludes zero while the large study's pins the effect down tightly. Sample size changed the precision, not the truth.

11. Multiple Confidence Levels

```
# Load data
opinion_data <- read_csv("data/public_opinion.csv")

# Fit model
model_opinion <- lm(policy_support ~ age + education_years, data = opinion_data)

# Different CIs for age coefficient
confint(model_opinion, parm = "age", level = 0.90)
```

```
          5 %      95 %
age 0.2912614 0.4970853
```

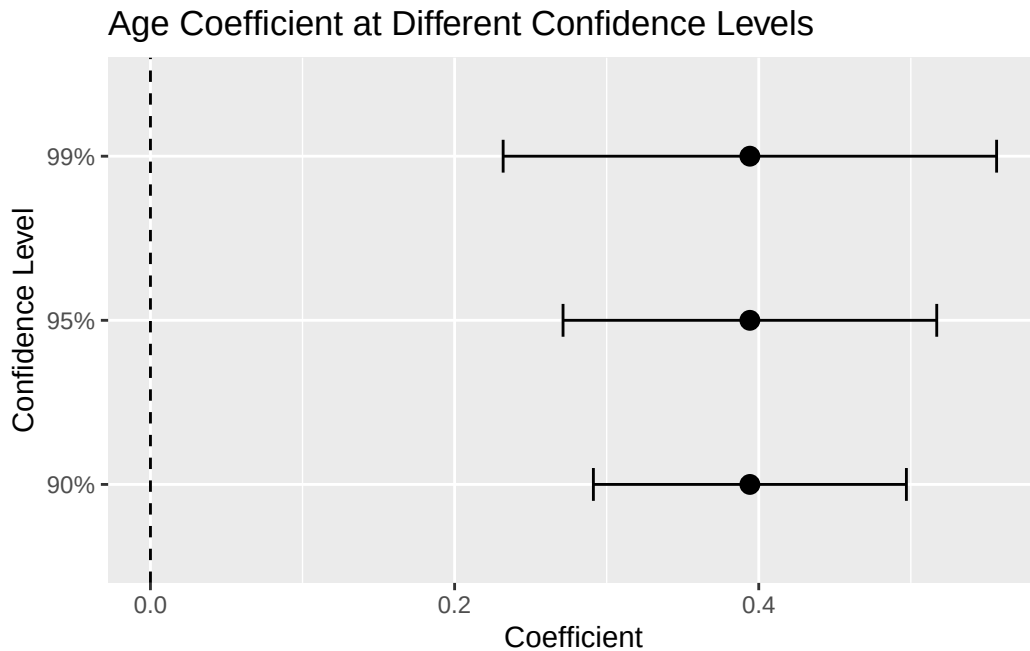
```
confint(model_opinion, parm = "age", level = 0.95)
```

```
      2.5 %    97.5 %  
age 0.2713075 0.5170391
```

```
confint(model_opinion, parm = "age", level = 0.99)
```

```
      0.5 %    99.5 %  
age 0.2319245 0.5564221
```

```
# Visualize  
tibble(  
  level = c("90%", "95%", "99%"),  
  lower = c(confint(model_opinion, parm = "age", level = 0.90)[1],  
            confint(model_opinion, parm = "age", level = 0.95)[1],  
            confint(model_opinion, parm = "age", level = 0.99)[1]),  
  upper = c(confint(model_opinion, parm = "age", level = 0.90)[2],  
            confint(model_opinion, parm = "age", level = 0.95)[2],  
            confint(model_opinion, parm = "age", level = 0.99)[2]),  
  estimate = coef(model_opinion)["age"]  
) %>%  
  ggplot(aes(x = estimate, y = level)) +  
  geom_vline(xintercept = 0, linetype = "dashed") +  
  geom_point(size = 3) +  
  geom_errorbarh(aes(xmin = lower, xmax = upper), height = 0.2) +  
  labs(title = "Age Coefficient at Different Confidence Levels",  
       x = "Coefficient", y = "Confidence Level")
```

Trade-off: Higher confidence requires wider intervals. Use 99% when you need high certainty; 95% is standard for most research.

12. Writing Practice

```
# Using education model
summary(model_edu)
```

Call:

```
lm(formula = test_score ~ spending_per_student, data = education_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-15.2338	-5.4363	-0.7018	4.6481	26.2830

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	6.584e+01	3.581e+00	18.386	< 2e-16 ***
spending_per_student	1.362e-03	2.849e-04	4.779	6.19e-06 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7.762 on 98 degrees of freedom

Multiple R-squared: 0.189, Adjusted R-squared: 0.1807

F-statistic: 22.84 on 1 and 98 DF, p-value: 6.189e-06

```
confint(model_edu)
```

	2.5 %	97.5 %
(Intercept)	58.733710901	72.946159919
spending_per_student	0.000796155	0.001926812

Complete write-up:

“We examined the relationship between per-student spending and test scores across 100 school districts. Education spending showed a significant positive relationship with test scores ($= 0.00136$, 95% CI: [0.00080, 0.00193], $t(98) = 4.78$, $p < 0.001$). Each additional \$1,000 in per-student spending was associated with a 1.36-point increase in average test scores. The confidence interval spans 0.8 to 1.93 points per \$1,000, so while the direction of the effect is clear, its magnitude is estimated only roughly. In practical terms, a \$3,000 increase in spending — about one standard deviation in this sample — would predict roughly a 4.1-point gain in test scores, which is meaningful but modest against a mean score near 83. These results suggest that spending matters, but that other factors likely play larger roles in determining educational outcomes.”

13. Simulation: Understanding “95% Confident”

```
# Function to run one study
run_study <- function(n = 50) {
  data <- tibble(
    x = rnorm(n, 50, 15),
    y = 20 + 0.04 * x + rnorm(n, 0, 8)
  )
  model <- lm(y ~ x, data = data)
  ci <- confint(model)[2, ]
  tibble(lower = ci[1], upper = ci[2], estimate = coef(model)[2])
}

# Run simulation
set.seed(202)
```

```

many_studies <- map_dfr(1:100, ~run_study(), .id = "study")

# Count how many contain true value
true_beta <- 0.04
many_studies %>%
  mutate(contains_true = lower <= true_beta & upper >= true_beta) %>%
  summarise(n_contain = sum(contains_true), pct = mean(contains_true))

```

```

# A tibble: 1 x 2
  n_contain  pct
    <int> <dbl>
1         96 0.96

```

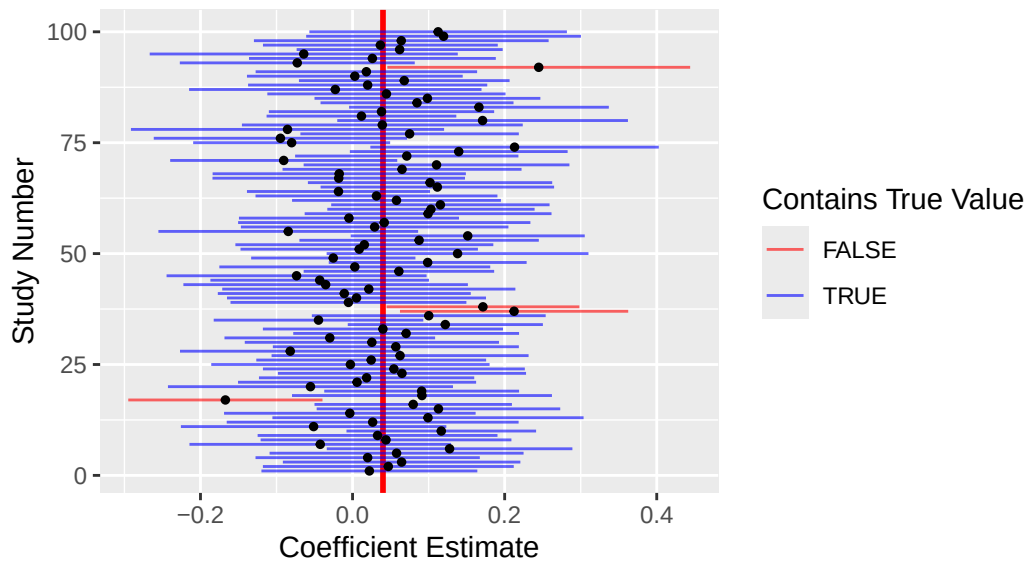
```

# Visualize
many_studies %>%
  mutate(
    study_num = as.numeric(study),
    contains_true = lower <= true_beta & upper >= true_beta
  ) %>%
  ggplot(aes(y = study_num)) +
  geom_vline(xintercept = true_beta, color = "red", size = 1) +
  geom_segment(aes(x = lower, xend = upper, yend = study_num,
    color = contains_true), alpha = 0.6) +
  geom_point(aes(x = estimate), size = 1) +
  scale_color_manual(values = c("red", "blue")) +
  labs(title = "100 Simulated 95% Confidence Intervals",
    subtitle = "Red line shows true value (beta = 0.04)",
    x = "Coefficient Estimate", y = "Study Number",
    color = "Contains True Value")

```

100 Simulated 95% Confidence Intervals

Red line shows true value (beta = 0.04)



Interpretation: Approximately 95 out of 100 CIs contain the true value. This demonstrates what “95% confident” means: if we repeated the study many times, 95% of our CIs would capture the true parameter.

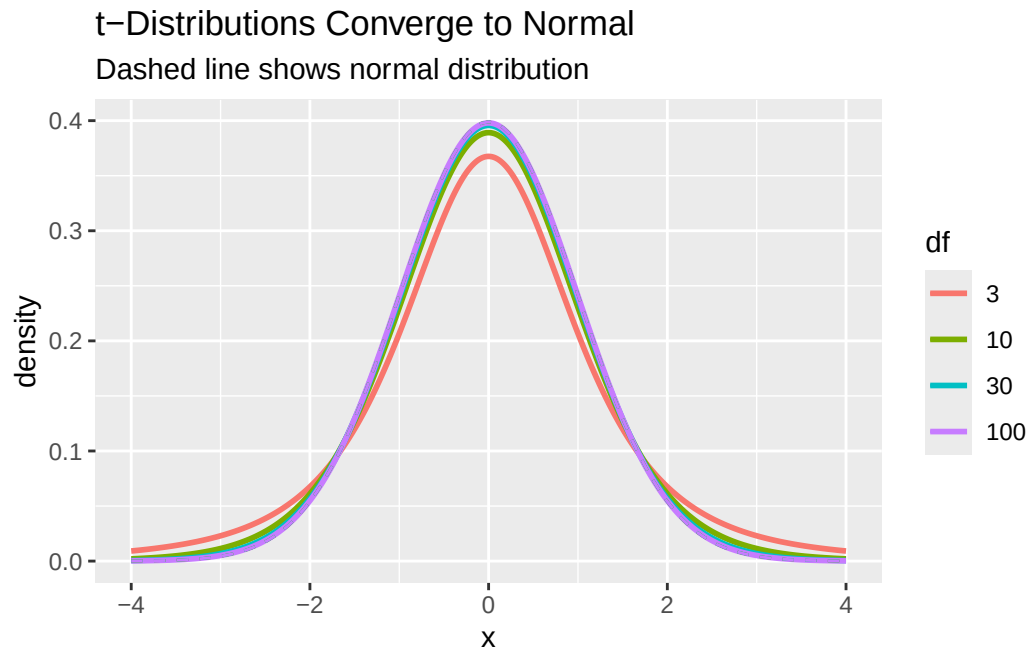
14. Critical Values Practice

```
# Table of critical values
expand_grid(
  n = c(10, 30, 100, 1000),
  level = c(0.90, 0.95, 0.99)
) %>%
  mutate(
    df = n - 1,
    prob = (1 + level) / 2,
    critical_t = qt(prob, df)
  )
```

```
# A tibble: 12 x 5
      n level  df prob critical_t
<dbl> <dbl> <dbl> <dbl>      <dbl>
1     10  0.9    9 0.95      1.83
```

2	10	0.95	9	0.975	2.26
3	10	0.99	9	0.995	3.25
4	30	0.9	29	0.95	1.70
5	30	0.95	29	0.975	2.05
6	30	0.99	29	0.995	2.76
7	100	0.9	99	0.95	1.66
8	100	0.95	99	0.975	1.98
9	100	0.99	99	0.995	2.63
10	1000	0.9	999	0.95	1.65
11	1000	0.95	999	0.975	1.96
12	1000	0.99	999	0.995	2.58

```
# Plot t-distributions
x_vals <- seq(-4, 4, by = 0.01)
tibble(
  x = rep(x_vals, 4),
  df = rep(c(3, 10, 30, 100), each = length(x_vals)),
  density = dt(rep(x_vals, 4), rep(c(3, 10, 30, 100), each = length(x_vals)))
) %>%
  ggplot(aes(x = x, y = density, color = factor(df))) +
  geom_line(size = 1) +
  stat_function(fun = dnorm, linetype = "dashed") +
  labs(title = "t-Distributions Converge to Normal",
        subtitle = "Dashed line shows normal distribution",
        color = "df")
```



15. Common Pitfalls Investigation

```
# Load data
large_sample <- read_csv("data/large_sample_study.csv")

# Fit model
large_model <- lm(outcome ~ intervention, data = large_sample)
summary(large_model)
```

Call:

```
lm(formula = outcome ~ intervention, data = large_sample)
```

Residuals:

Min	1Q	Median	3Q	Max
-11.7597	-2.0156	0.0003	2.0316	11.2057

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	50.01975	0.04255	1175.476	< 2e-16 ***
intervention	0.28452	0.06009	4.735	2.22e-06 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.004 on 9998 degrees of freedom

Multiple R-squared: 0.002238, Adjusted R-squared: 0.002138

F-statistic: 22.42 on 1 and 9998 DF, p-value: 2.22e-06

```
confint(large_model)
```

```
                2.5 %      97.5 %  
(Intercept) 49.9363333 50.1031572  
intervention  0.1667382  0.4023098
```

```
# Effect size
```

```
coef(large_model)["intervention"]
```

```
intervention  
      0.284524
```

```
# CI width
```

```
ci <- confint(large_model)["intervention", ]  
ci[2] - ci[1]
```

```
      97.5 %  
0.2355716
```

Interpretation: The intervention has a highly significant effect ($\beta = 0.285$, 95% CI: [0.167, 0.402], $t = 4.74$, $p < 0.001$). However, it represents only a 0.57% change from the control baseline of 50 on a 0-100 scale — negligible in practical terms. The CI is very narrow because $n = 10,000$, which is exactly what lets the study detect a trivially small effect. **Lesson:** Statistical significance does not equal practical importance, especially with large samples.

16. Complete Analysis Project

Students should choose a substantive question and demonstrate:

1. Appropriate model specification
2. Correct CI calculation
3. Clear interpretation of results
4. Discussion of precision (CI width)
5. Consideration of both statistical and practical significance
6. Proper write-up following academic standards